Reducing sourcing costs for caseworkers

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Abstract

In late 2024, the French public employment service introduced a new AI tool to support caseworkers dedicated to assisting client recruiters in their hiring decisions. This tool was designed to reduce the time spent by caseworkers on sourcing candidates. We leverage the randomised implementation of this intervention to measure its impact on job-filling probabilities. We find that the intervention successfully increased the probability of filling posted jobs by between 10 and 15 per cent. While the intervention was intended solely to facilitate the sourcing of job seekers who had not initially applied for the job, we find that the increase in hires comes from both job seekers who were sourced and job seekers who had applied on their own (but had to be screened by caseworkers). Our interpretation is that the AI tool freed up caseworkers' time, allowing them to devote more time to screening applications (and other tasks).

Keywords: job search, hiring, public employment service

JEL Codes: J2, J6, J7

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1 Introduction

Most countries experience both persistent and sizeable levels of unemployment and unfilled vacancies. The co-existence of these two phenomena can be explained either by the existence of mismatch, whether geographic or skill-related, or by frictions in the matching process. A large strand of the literature has explored and broadly rejected the idea that mismatch could be a sizeable driver of persistent unemployment and hiring difficulties, leaving frictions as the most likely determinant. Frictions themselves can be very diverse. Some frictions are rooted in a lack of information on the part of job seekers about existing vacancies, or about whether they are a good fit for these vacancies. Public Employment Services deploy many services (e.g. job-search assistance) to guide job seekers in their search. Policies as well as interventions aiming to provide more information to caseworkers have been evaluated by a large number of papers, and are shown to improve re-employment outcomes at the micro level Card et al. (2018); Altmann et al. (2018); Schiprowski (2020); Belot et al. (2022); Altmann et al. (2023); Le Barbanchon et al. (2023); Behaghel et al. (2024); Cederlöf et al. (2025), but questions remain as to whether they are effective in solving imbalances between labour demand and labour supply at the macro level Crépon et al. (2013); Gautier et al. (2018); Cheung et al. (2025).

In this paper, we measure the impact of an intervention that aims to support caseworkers in providing services to recruiters during the hiring process. We believe this kind of intervention is relevant for several reasons. First, because the demand side is often the short side of labour markets, alleviating frictions stemming from information imperfections faced by employers may be more promising in reducing labour market imbalances. Second, the literature provides much less evidence on the impact of interventions aiming to improve the information set of employers than on interventions targeting job seekers (with the notable exceptions of Pallais (2014); Algan et al. (2022) in developed countries and Carranza et al. (2022); Abel et al. (2020) in developing countries). Third, the existence of duration dependence (e.g. Kroft et al. (2013)) or discrimination in labour markets can be interpreted as indirect evidence that employers find it very costly to screen or source relevant candidates. Fourth, public employment services in several countries are currently ramping up the provision of services to recruiters. It is therefore worth understanding how this support can be provided in the most effective manner.

This paper takes place in France, where the public employment service, France Travail, has already deployed a fairly comprehensive set of services to firms. Some caseworkers are specifically dedicated to supporting employers. Employers posting a job advertisement at France Travail can access a caseworker who helps them to both (i) screen applications that job seekers submit on their own and (ii) source job seekers who have not applied but would be relevant for the job. At the end of the process, after checking both the relevance of candidates and their interest in the job, France Travail caseworkers provide recruiters with a short list of candidates. These activities are time-consuming. The intervention we study in this paper aims to support the work of caseworkers when they perform the sourcing task: instead of having to call job seekers to check their availability, their interest, and their fit, caseworkers can deploy an AI chatbot service, *MatchFT*, which engages in an SMS conversation with job seekers selected by caseworkers, answers their queries, and records both their potential interest and their fit.

To evaluate the impact of this intervention, France Travail designed a randomised controlled trial. Employers were randomised into two groups: (i) the treatment group, which benefits from the *MatchFT* chatbot service, and (ii) the control group, which does not and remains under business as usual. Caseworkers are free to use the service or not. Take-up was far from systematic: we observe that *MatchFT* was used for only between 30% and 40% of eligible jobs. Our main outcome of interest is the probability that a posted job is filled by an unemployed job seeker registered at France Travail. Treated firms hired 10% to 15% more unemployed job seekers than control ones, an impact that is quantitatively large considering the take-up rate.

Our data allow us to know whether hired workers were screened (i.e. they applied to the job on their own and were shortlisted by caseworkers) or sourced (i.e. they had not applied before caseworkers contacted them about the job). While the intervention only reduces the cost of sourcing, we find that the increase in the number of hires comes half from sourced workers and half from screened workers. One way to interpret this result is that *MatchFT* freed up time that caseworkers used for other tasks, in particular screening candidates for the same job advertisement.

We explore other sources of spillovers by leveraging the fact that, treatment being randomly assigned at the employer level, the share of job advertisements eligible to *MatchFT* that a caseworker had to handle was as good as random. Interestingly, we find that an increase in the share of treated job advertisements is correlated with an increase in the number of hires experienced by control job advertisements. This means that control job advertisements benefitted from being handled by caseworkers who had more time freed up by the intervention. Another consequence of this result is that the SUTVA is unlikely to hold, and that the treatment effect we measure when comparing treated and control job advertisements is likely to be a lower bound of the true impact.

Our paper contributes to four strands of the literature. First, the literature that aims to measure and understand the consequences of information frictions faced by firms when hiring Pallais (2014); Abel et al. (2020); Carranza et al. (2022); Algan et al. (2022). Our first contribution is to reassert the importance of information frictions faced by firms: tools that improve the performance of caseworkers in charge in supporting employers have the potential to improve hiring outcomes.

Second, the literature that aims to measure the importance of caseworkers in the matching process between labour supply and labour demand Schiprowski (2020); Cederlöf et al. (2025). We contribute by showing that services provided by caseworkers can be enhanced and effectively complemented by AI tools.

Third, the literature on how artificial intelligence can be leveraged to help match supply and demand in labour markets (and other markets) Bied et al. (2023); Le Barbanchon et al. (2023); Behaghel et al. (2024). While most papers in this literature evaluate the impact of AI-based recommender systems that provide recommendations to job seekers, this paper shows that AI tools can also be used to save human time and to substitute for routine tasks (such as scheduling meetings or extracting basic information).

Fourth, the literature about the performance of HR officers and civil servants. A strand of literature has looked into the influence of selection devices, the influence of managers or incentives to increase performance among civil servants Dal Bó et al. (2013); Biasi (2021); Muñoz and Prem (2024); Muñoz and Otero (2025). Hoffman et al. (2018) shows that the introduction of a guiding tool for HR officers improves their decisions. Our contribution is to show that it is possible to use AI tools to effectively complement workers' actions, freeing up their time on some tasks to increase their overall performance.

The remainder is organised as follows. We introduce the setting and the experimental design in the next section. Section 3 presents the data and descriptive statistics. In Section 4, we explain the methodology and provide the main results on the impact of the intervention on hiring outcomes. In Section 5, we explore the mechanisms.

2 Setting

2.1 Institutional setting

France Travail is the French public employment service. Its main mission is to help registered unemployed job seekers find a job and to help employers match with unemployed job seekers. France Travail maintains a job portal, `francetravail.fr`, which is the largest job board in the country, with typically around 1 million posted jobs on a given day.

As is the case in many settings, the most common way in which matching takes place is asymmetric: employers post jobs on the job board, job seekers browse the platform and apply to jobs (applications are initiated within the platform but typically finalised on firms' own websites), and employers then choose whom to hire among the applications they receive.

France Travail has a dedicated team of caseworkers specialised in supporting employers. Firms that request it can receive support from caseworkers throughout the hiring process, as described in Algan et al. (2022). Most importantly, once they have posted a job advertisement, employers can request assistance from France Travail to help them find suitable candidates for the job; we refer to this support track as "preselection" throughout the text. In this case, caseworkers perform two types of actions:

1. *Screening.* Job seekers can apply to this job advertisement on their own, but the quality of applications may be heterogeneous. One service provided by caseworkers is to screen the pool of applications and shortlist those that are a good fit for the job.

2. *Sourcing.* In addition to screening, caseworkers may decide to extend the list by sourcing candidates from the profile bank (i.e., all job seekers who have agreed to be part of this bank). Typically, caseworkers first use a matching tool that identifies job seekers in the bank who correspond to the main characteristics of the job advertisement (e.g., occupation, location, experience). They then contact, by email or by phone, the candidates whom they consider to be a good fit.

Caseworkers then compile the list of job seekers they have screened and sourced and hand it over to the recruiter.

2.2 Intervention

Late 2024, France Travail started to roll out *MatchFT*, a new tool made available to caseworkers to help them source job seekers for jobs in preselection. This tool consists in an AI-driven SMS chatbot that is designed to automatically probe the interest and the availability of a job seeker for a given job, assess how well they would fit the job, and answer questions that job seekers may have about the contents of the job.

In practice, when a caseworker uses the matching tool to get a list of potential suitable candidates in the bank of profiles, the tool provides a button in front of the name of each candidate. When the caseworker presses a button, an SMS conversation between the chatbot and the candidate would start. The tool then summarises the content of the conversation (e.g., whether the candidate is interested or not, whether they are skilled for the job or not) within the usual dashboard of caseworkers.

The main objective of the tool is to save caseworkers' time, by substituting potentially time-consuming email exchanges or phone conversations.

2.3 Experimental protocol

The random assignment into treatment and control was conducted by France Travail using a stratified sampling procedure. The sampling frame consisted of all registered establishments that had published at least one job offer since 2019. The primary stratification variable was the number of job offers published in 2023. France Travail first estimated the expected number of job postings in 2023 using regression models based on establishment-level characteristics. For establishments with complete information, the predicted value was obtained from a linear regression estimating the probability of being active (i.e., publishing at least one ad in 2023), using the following covariates: establishment age, sector of activity, number of job offers published in 2022, number of job offers published between July and December 2022, region, and employment size category. For establishments with missing values, a feature of the employment size and region variables, France Travail ran complementary regressions excluding the missing variables.

For each regression – full information sample, missing establishment size, missing region – establishments were grouped into deciles based on the predicted number of postings from their respective regression. The random assignment was carried out within each decile stratum. Establishments with missing values were not excluded but were stratified separately, according to the deciles derived from the corresponding complementary regression.

3 Data

3.1 Data sources

3.1.1 France Travail administrative data

We rely on job advertisements posted at France Travail. For each job advertisement, we observe the occupation, the place of work, the employer, the date of publication, and the date on which the advertisement is closed. Every time a job ad is posted at France Travail, a proprietary algorithm classifies it as either easy or hard to fill, based on its characteristics (e.g., occupation, location). We use this variable (which we sometimes refer to as *IEE score*) to study the heterogeneity of the treatment effect with respect to the extent of hiring difficulty.

This paper focuses on job advertisements that employers have chosen to place in preselection (one of the hiring assistance services provided by France Travail). In the administrative data, we observe the date when the job advertisement is placed in preselection, as well as the identifier of the caseworker (and the agency) handling the job advertisement.

For jobs in preselection, we observe which job seekers have been shortlisted by caseworkers (and hence delivered to recruiters). We also observe applications made by job seekers on their own, whether they have been shortlisted or screened out by caseworkers. For sourcing in the control group (business as usual), we do not observe the universe of job seekers who have been sourced if they are not shortlisted. For sourcing in the treatment group, we observe when job seekers have been contacted via the *MatchFT* chatbot, even when they are not shortlisted subsequently.

3.1.2 Déclaration Sociale Nominative (DSN)

We use administrative employer–employee data drawn from the *Déclaration Sociale Nominative* (DSN), the French mandatory monthly social declaration system. The DSN requires all private-sector employers to report, on a continuous basis, detailed information on their employees and payroll to the social security administration. DSN declarations are collected and processed by French social security institutions.

The DSN provides exhaustive, individual-level information on employment relationships, including gross and net earnings, hours worked, employment contracts, occupational status, firm identifiers, and key demographic characteristics of workers. Data are reported at a monthly frequency and cover virtually the entire population of private-sector employees in France.

The DSN is transmitted to France Travail and matched, at the individual and employer (establishment) level, with the job advertisement data and the application data. We use it to detect matches between a job seeker and an employer following an application.

3.2 Outcomes

3.2.1 Job filling

The main outcome of interest is the extent to which a posted job in preselection is filled, and in particular, filled by a registered unemployed job seeker. By definition, jobs in the preselection track can only be filled (within the France Travail platform) by applicants who are shortlisted by caseworkers, as employers do not observe other potential France Travail applicants. We construct our main outcome at the job-advertisement level by linking caseworker shortlists to hires using administrative data from the DSN. For each job advertisement, we identify hires of candidates who had been shortlisted by caseworkers. We match DSN work contracts to the preselection shortlist using establishment and individual identifiers. We restrict attention to hires with contract start dates on or after the date of shortlist creation. For each advertisement, we then count the number of hires that begin within a given time window (e.g., 14 or 30 days) after the start of the preselection process. This outcome includes all hires corresponding to shortlisted candidates, regardless of whether they came from the *MatchFT* tool.

3.2.2 Shortlisted candidates

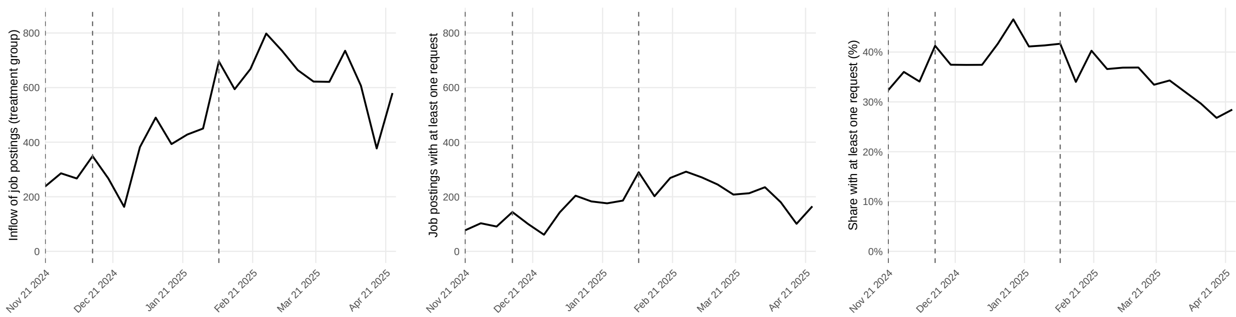
To shed light on the mechanisms behind the impact of the intervention, we use data on shortlisted candidates. First, we study the impact on the number of candidates shortlisted. Second, we investigate whether shortlisted candidates came from the screening process (i.e., they applied to the job on their own and were selected by caseworkers) or the sourcing process (i.e., they had applied to the job but were contacted by caseworkers and nudged into applying).

3.3 Descriptive statistics

3.3.1 Experiment roll-out and take-up

Figure 1 shows the evolution of take-up of the *MatchFT* tool. Panel (a) shows the total weekly number of job advertisements in preselection that are assigned to the treatment, i.e. the eligible population. This number increases from around 300 in late 2024 to around 700 in February–March 2025, largely reflecting the increase in the number of agencies involved in the experiment. Panel (b) shows the weekly number of job advertisements with at least one *MatchFT* request, which lies between 100 and 300. Panel (c) shows the ratio between the two panels: the share of eligible advertisements for which we observe at least one *MatchFT* request. This ratio is fairly stable at around 40%, before steadily decreasing to 30% by the end of the period.

Figure 1: Evolution of take-up of the *MatchFT* service among treated job ads.



(a) Weekly number of ads posted by treated firms.

(b) Number of job ads with at least one *MatchFT* request.

(c) Share of treated ads that receive at least one *MatchFT* request.

Notes: The date refers to the day at which the preselection process started for the ad. Vertical dashed lines mark the roll-out dates: November 21, 2024 (first roll-out in a subset of agencies), December 12, 2024 (roll-out in the remainder of agencies), and February 6, 2025 (increase of treatment share to 75%).

3.3.2 Sample description and balancing checks

Table 1 presents summary statistics for the control group in terms of baseline characteristics. All variables are measured over the 12-week period preceding treatment assignment.

To check the validity of the randomisation process, we also show baseline differences between treated and control establishments. Differences between treated and control groups are estimated using the regression specification in Equation (1).

$$Y_{ies} = \alpha + \beta \cdot \text{Treated}_i + \delta_e + \lambda_s + \varepsilon_{ies}, \tag{1}$$

where Y_{ies} denotes the cumulative outcome for establishment i in subexperiment e and strata s; Treated_i is an indicator for whether the establishment was treated in that subexperiment; δ_e and λ_s are subexperiment and strata fixed effects, respectively; and ε_{ies} is the error term.

3.3.3 Usage of the *MatchFT* tool

In Table 2, we investigate how caseworkers used the *MatchFT* tool when they used it. In total, 110,137 chatbot interactions were initiated across 4,151 job advertisements. Condi-

Table 1: Balance check and descriptive statistics (pre-experiment period)

	Always control (N = 2141) Mean	Always treated (N = 4308) Diff (p)	Second roll-out (N = 2158) Diff (p)
Establishment size			
10 or less	0.449	-0.000 (0.976)	0.003 (0.822)
11–25	0.138	0.003 (0.753)	0.012 (0.267)
26–50	0.156	-0.009 (0.366)	0.009 (0.423)
More than 50	0.183	-0.001 (0.950)	-0.023** (0.048)
Missing	0.075	0.007 (0.340)	-0.002 (0.850)
Establishment industry			
Construction	0.069	0.001 (0.857)	-0.010 (0.168)
Manufacturing	0.116	0.006 (0.464)	0.001 (0.924)
Commerce	0.156	-0.005 (0.575)	0.004 (0.728)
Services	0.150	0.005 (0.606)	0.010 (0.369)
Health	0.140	-0.004 (0.654)	-0.002 (0.848)
Ad posting & caseworker referrals			
Ads posted	13.470	-0.216 (0.896)	-1.697 (0.308)
Ads posted (pre-sel.)	2.259	-0.113 (0.458)	-0.127 (0.429)
Candidates shortlisted	17.028	-1.311 (0.236)	-1.335 (0.265)
Hiring			
Hires via platform	0.708	0.010 (0.882)	-0.032 (0.657)

tional on sending at least one *MatchFT* request for a given job advertisement, caseworkers used the tool on 27 job seekers on average.

Caseworkers can choose which criteria they want the chatbot to assess. Interest in the job is compulsory (present in 100% of requests), but availability (83%), adequacy with tasks (61%), possible mobility (61%), and relevant experience (57%) were also frequently assessed by caseworkers.

Did job seekers engage with the chatbot? Ultimately, only 5% of requests concerning interest resulted in a positive response (i.e., job seekers did not ignore the tool and responded positively). This implies that the tool generated, on average, 1.3 interested candidates per job advertisement when it was used.

Caseworkers appear to have taken the information provided by *MatchFT* seriously. Forty-three per cent of candidates who indicated interest in the job via the chatbot were ultimately shortlisted by caseworkers, compared with 2% of candidates who responded negatively or did not respond (and may have been contacted independently by caseworkers).

Table 2: Usage of *MatchFT*

Criteria	Asked in N ads	Share	Number requests	Positive answers	Shortlist among pos. answers	Shortlist among neg. answers
Interest	4151	100.0%	110137	5.0%	42.7%	1.5%
Availability	3464	83.4%	88442	4.1%	50.3%	1.8%
Tasks	2631	63.4%	68348	3.7%	54.0%	1.9%
Mobility	2530	60.9%	68253	3.6%	50.6%	1.7%
Experience	2380	57.3%	63906	3.4%	52.3%	1.9%
Working hours	1596	38.4%	35458	3.6%	49.4%	2.1%
Wage expectations	1420	34.2%	33687	3.6%	48.1%	2.0%

Notes: Summary statistics for job postings in treatment group with at least one *MatchFT* request. No answers ('Answer pending') are counted as negative answers.

3.4 Screening vs. sourcing

Table 3 compares the number of shortlisted candidates per posting for all postings with those for which we observe at least one *MatchFT* request. This comparison cannot be interpreted causally, as the two sets of postings have no reason to be similar.

In the sample of all preselection postings, we observe that caseworkers shortlist an average of eight candidates, roughly six coming from screening and two from sourcing. Postings with at least one *MatchFT* request have a slightly longer shortlist (around ten candidates) and a higher share of shortlisted candidates coming from sourcing.

Table 3: Number of shortlisted candidates per posting

	All postings		Postings with MatchFT req.	
	Mean	Median	Mean	Median
Total candidates per posting	8.2	5.0	9.5	7.0
Shortlisted via screening	5.8	3.0	6.1	7.0
Shortlisted via sourcing	2.3	1.0	3.5	3.0
- out of which after MatchFT request	0.4	0.0	2.0	1.0

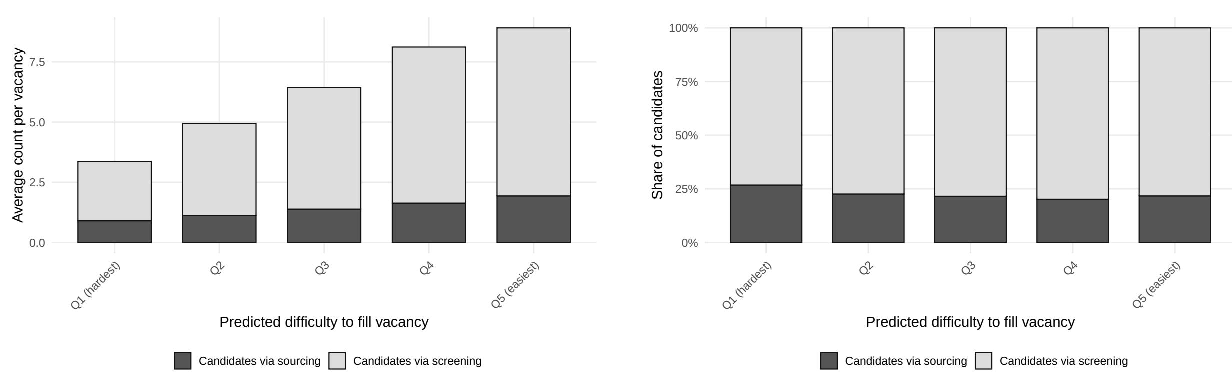
Notes: Summary statistics for job postings in treatment group with at least one referral

Figure 2 shows how the number and composition of shortlisted candidates vary across quintiles of the IEE score, which rates how easy a job is to fill. First, we assess the relevance of this score. Jobs in the 20% hardest to fill receive, on average, three shortlisted candidates, while those in the 20% easiest to fill receive an average of nine.

While the total number of shortlisted candidates varies, the share coming from sourcing versus screening does not, and remains between 20% and 25% across all quintiles. Is this result counter-intuitive? One might have expected that, for vacancies that are easier to fill, there are enough job seekers who apply on their own, so that caseworkers do not need to engage in sourcing. Conversely, for jobs that are hard to fill, there are fewer applicants, and caseworkers may want to compensate by performing more sourcing. This does not appear to be the case, possibly because the returns to sourcing also decrease with the difficulty of filling the vacancy.

Figure 3 shows how the number and composition of hired unemployed job seekers vary with how difficult a job is to fill. Again, the number of hires increases with how easy the job is to fill, which is intuitive. In addition, the share of hired job seekers who were screened versus sourced is constant across quintiles. Interestingly, the share of sourced job seekers among those hired is around 50%, while sourced shortlisted applicants represent only about 25%. This suggests that, all else equal, sourced workers are likely to be of higher quality than screened ones.[1]

Figure 2: Composition of shortlisted candidates by predicted vacancy filling difficulty (control group).



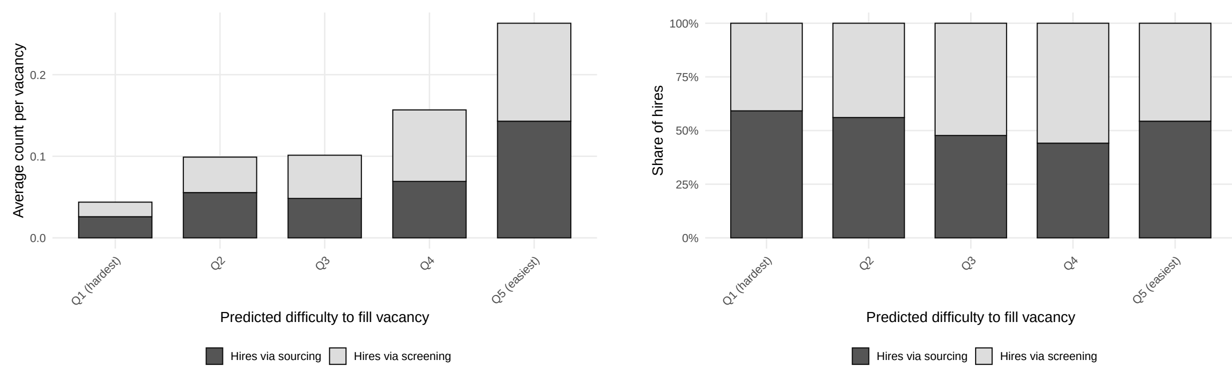
(a) Average number of candidates per vacancy.

(b) Share of candidate types.

Notes: Sample restricted to control group vacancies with non-missing IEE score. Quintiles are based on the IEE score, a prediction of the probability of filling the vacancy within 30 days computed by France Travail, where Q1 contains the hardest-to-fill vacancies and Q5 the easiest. Candidates via sourcing are caseworker-initiated contacts or applications following a *MatchFT* recommendation. Candidates via screening are job seeker applications approved as referrals by the caseworker, excluding those triggered by *MatchFT* recommendations. Counts are measured within 30 days of the vacancy entering the preselection process.

[1] An alternative explanation, which we cannot exclude, is that this difference comes from workers' behaviour, whereby sourced workers are more likely to accept the employer's offer.

Figure 3: Composition of hires via portal by predicted vacancy filling difficulty (control group).



(a) Average number of hires via platform within 30 days of recruitment start.

(b) Share of hire types.

Notes: Sample restricted to control group vacancies with non-missing IEE score. Quintiles are based on the IEE score, a prediction of the probability of filling the vacancy within 30 days computed by France Travail, where Q1 contains the hardest-to-fill vacancies and Q5 the easiest. Hires via sourcing are hires of candidates referred through caseworker-initiated contacts or *MatchFT* recommendations. Hires via screening are hires of candidates referred through caseworker-approved job seeker applications. Hires are measured within 45 days of the vacancy entering the preselection process.

4 The impact of reducing caseworkers' screening costs on hiring outcomes

4.1 Methodology

Our baseline analysis is conducted at the advertisement level. We consider all ads that entered the preselection stage and define the outcome variable as an indicator for whether a hire was made through a referral within a given number of days after the ad was posted.

A key empirical challenge arises from the fact that both ad-level filling rates and the rollout of the treatment vary over time. This raises concerns about potential confounding due to secular trends or shifts in aggregate referral activity. Moreover, treatment was implemented in a staggered fashion across strata, following a redraw for the increased roll-out on February 6 2025.

To address these concerns, we control for baseline differences in fill rates across strata and over time. Specifically, we include fixed effects for each stratum interacted with an indicator for the period following the main increase in treatment exposure, which occurred on February 6, 2025. This strategy allows us to flexibly account for time-varying differences across strata unrelated to the treatment.

Formally, we estimate the following specification:

$$Y_i = \beta D_{f(i)} + \theta_{s(i)} \times \infty \left[t(i) \geq \text{Feb 6, 2025} \right] + \varepsilon_i,$$

where Y_i is the binary outcome for ad i, $D_{f(i)}$ is a treatment indicator for the firm or establishment associated with ad i, $\theta_{s(i)}$ denotes fixed effects for the stratum s to which ad i belongs, and the interaction term captures post-period variation in baseline outcomes across strata. Standard errors are clustered at the firm or establishment level.

When estimating treatment effects, we account for the stratified random assignment by including strata fixed effects. However, conventional standard errors may be conservative in the context of stratified random assignment when the treatment probability within strata deviates from 0.5 (Bugni et al., 2018). To address this, we apply the correction proposed by Bugni et al. (2018) using the `sreg` package for `R` (Trifonov et al., 2024). To account for the fact that treatment was assigned at the establishment level and some establishments switch treatment status on February 6, we aggregate outcomes at the establishment-by-subexperiment level. Specifically, we compute the average outcome for each establishment across all job postings, at each evaluation horizon, within each subexperiment. This avoids the need for clustered standard errors and ensures correct alignment between the unit of treatment assignment and the unit of observation in the estimation.

4.2 Impact on job-filling probability

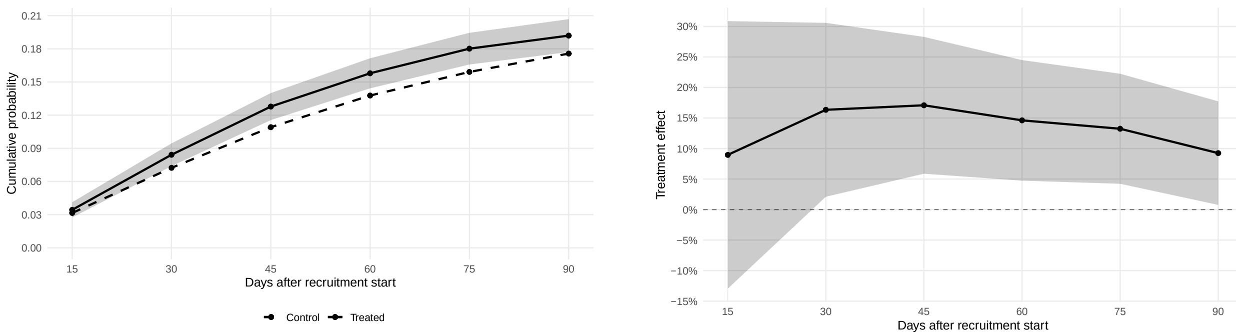
Figure 4 presents our main results on the treatment effect of the matching tool on vacancy filling through the PES. Panel (a) displays the cumulative probability of hiring a candidate within a given number of days after the vacancy enters preselection mode. By day 45, treated vacancies have a hiring probability of approximately 13%, compared to 10% for control vacancies.

Panel (b) expresses these effects as a percentage of the control mean. The treatment effect peaks at approximately 15% around day 45, after which control vacancies begin to catch up. This pattern suggests that the matching tool generates both speed and level effects: the larger relative effect at earlier horizons reflects faster hiring for treated vacancies, while the persistent and statistically significant effect at 90 days indicates that some of these additional hires might not have occurred absent treatment. Table 4 reports the corresponding estimates.

Our results are robust to alternative specifications. Tables A1–A3 in the Appendix show that the estimates are stable when using only strata fixed effects or when including additional controls for the number of positions and pre-experiment hiring history. Table A4 presents Poisson estimates for the number of hires, which yield similar conclusions. Table A5 shows that results also hold when using an alternative outcome measuring whether all advertised positions were filled.

To validate our experimental design, we conduct a placebo test using vacancies from November 2023 to April 2024, a year before the experiment began. We assign placebo treatment status based on establishments' eventual treatment assignment. Figure A1 and Table A7 in the Appendix show no significant differences between treatment and control groups in the pre-treatment period, confirming the absence of differential pre-trends.

Figure 4: Treatment Effect on Vacancy Filling



(a) Cumulative vacancy filling probability

(b) Treatment effect

Notes: Panel (a) shows the cumulative probability of filling a vacancy through the PES within the specified number of days after recruitment start, separately for treated and control vacancies. Panel (b) shows the treatment effect as a percentage of the control mean at each horizon. Shaded areas and error bars represent 95% confidence intervals.

Table 4: Treatment Effect on Vacancy Filling

	Vacancy filled					
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.0028 (0.0035)	0.0118** (0.0053)	0.0186*** (0.0062)	0.0201*** (0.0069)	0.0211*** (0.0073)	0.0162** (0.0076)
Strata $\times$ sub-experiment FE	X	X	X	X	X	X
Observations	10,201	10,201	10,201	10,201	10,201	10,201
Control mean	0.0315	0.0723	0.1091	0.1377	0.1591	0.1757

Notes: Results from horizon-specific linear regressions. Outcome: binary indicator of hire through the PES for the job posting. One observation is one establishment, if an establishment has posted more than one posting, the filling rates are averaged at the horizon level. Subexperiment denotes an indicator variable whether the preselection process for the vacancy started after Feb 6, 2025 when the treatment share was increased from 50 to 75%. The outcome is averaged at the establishment $\times$ subexperiment level and the standard errors are adjusted following **?**.

5 Mechanisms

5.1 Number of candidates shortlisted

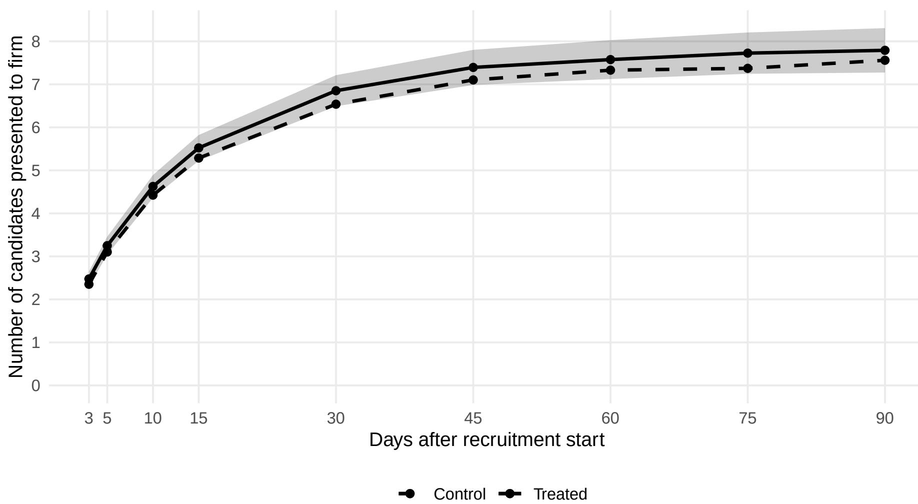
A natural first channel through which the matching tool could improve hiring outcomes is by increasing the number of candidates presented to employers. If caseworkers use the tool's recommendations to identify additional suitable candidates, one might expect treated vacancies to receive more referrals. Figure 5 shows that the matching tool has a minimal, statistically insignificant effect on the number of candidates presented to firms. Point estimates are small and indistinguishable from zero across all time horizons at the 5% level. By day 45, both treated and control vacancies receive approximately 7.1 candidates on average, with treatment effects of less than 5%. This null result on quantity is consistent with an institutional feature of France Travail's service model. Caseworkers typically set themselves a target of 5–10 candidates to present per vacancy, aiming to provide employers with a manageable shortlist rather than overwhelming them with applications (Algan et al., 2022).

The absence of a quantity effect, combined with the significant improvement in hiring outcomes documented in Figure 4, suggests that the matching tool operates primarily through a quality channel: caseworkers present candidates who are better matched to vacancies, rather than simply presenting more candidates. In the following subsections, we investigate how this quality improvement manifests across two distinct referral channels—caseworker-initiated sourcing versus screening of job seeker applications—and how the tool's effectiveness varies with the caseworkers workload and the vacancy's predicted filling difficulty. Table A8 in the Appendix reports the full regression results.

5.2 Sourcing vs. screening

The matching tool could improve hiring outcomes through two distinct channels.
Figure 6 decomposes the treatment effect by those two channels. Panels (a) and (b) focus on hires via sourcing, defined as caseworker-initiated referrals. The treatment effect on sourcing is positive and statistically significant at the 60 days horizon, with treated vacancies approximately 15% more likely to result in a hire through this channel. Panels (c) and (d) examine hires via screening, where the caseworker approves a job seeker's application. The treatment effect on screening is highest after 30 days and statistically significantly different from zero from day 30 throughout the whole analysis horizon. Table 5 reports the corresponding estimates.

Figure 5: Treatment Effect on Number of Candidates Shortlisted per Vacancy Posting



Notes: The figure shows the cumulative number of candidates presented to firms through the PES within the specified number of days after recruitment start, separately for treated and control vacancies. Estimates from Poisson regressions with strata × subexperiment and agency × subexperiment fixed effects. Shaded areas represent 95% confidence intervals computed using the delta method, clustered at the establisment level.

Table 5: Treatment effect by channel (sourcing vs. screening)

Panel A: Hires via sourcing

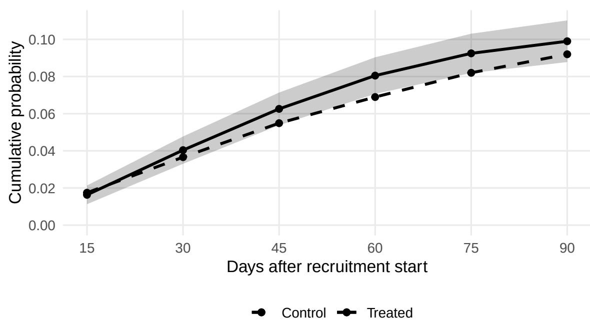
	Vacancy filled					
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	-0.0011 (0.0026)	0.0038 (0.0037)	0.0077* (0.0045)	0.0115** (0.0050)	0.0105* (0.0054)	0.0070 (0.0057)
Strata × sub-experiment FE	X	X	X	X	X	X
Observations	10,201	10,201	10,201	10,201	10,201	10,201
Control mean	0.0174	0.0366	0.0549	0.0689	0.0820	0.0920

Panel B: Hires via screening

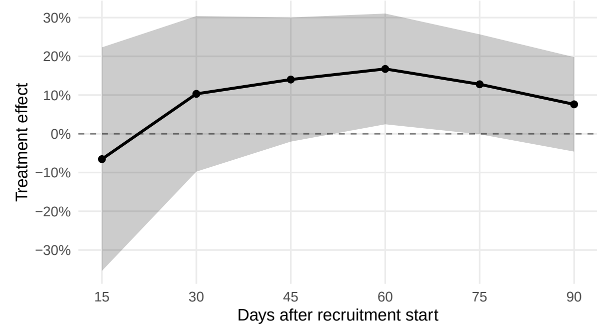
	Vacancy filled					
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.0038 (0.0025)	0.0092** (0.0039)	0.0121*** (0.0047)	0.0101* (0.0053)	0.0121** (0.0055)	0.0105* (0.0058)
Strata × sub-experiment FE	X	X	X	X	X	X
Observations	10,201	10,201	10,201	10,201	10,201	10,201
Control mean	0.0146	0.0368	0.0561	0.0723	0.0822	0.0903

Notes: Results from horizon-specific linear regressions. Panel A: outcome is a binary indicator of hire via sourcing (caseworker-initiated referral). Panel B: outcome is a binary indicator of hire via screening (caseworker-approved job seeker application). One observation is one establishment, if an establishment has posted more than one posting, the filling rates are averaged at the horizon level. Subexperiment denotes an indicator variable whether the preselection process for the vacancy started after Feb 6, 2025 when the treatment share was increased from 50 to 75%. The outcome is averaged at the establishment × subexperiment level and the standard errors are adjusted following Bugni et al. (2018).

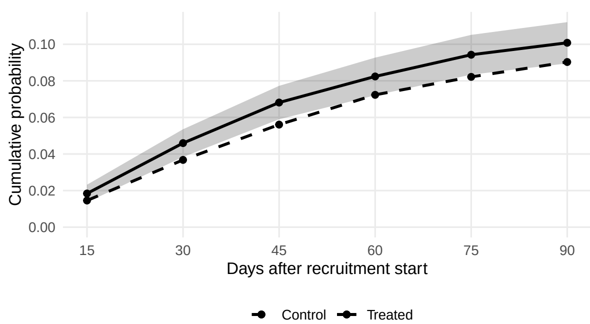
Figure 6: Treatment effect by channel (sourcing vs. screening)



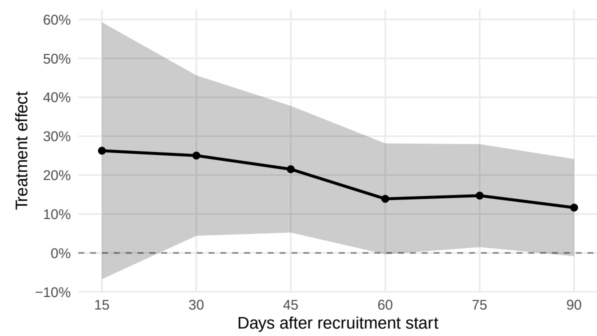
(a) Sourcing – Cumulative probability



(b) Sourcing – Treatment effect



(c) Screening – Cumulative probability



(d) Screening – Treatment effect

Notes: Panels (a) and (b) show treatment effects on hires via sourcing (caseworker-initiated referrals), while panels (c) and (d) show effects on hires via screening (caseworker-approved job seeker applications). Left panels display cumulative hiring probabilities for treated and control vacancies. Right panels display treatment effects as a percentage of the control mean. Shaded areas and error bars represent 95% confidence intervals.

5.3 Caseworker congestion and time gains

If the matching tool reduces the time caseworkers spend on sourcing for treated vacancies, this time savings may spill over to benefit other vacancies in their caseload. Conversely, the benefits of the matching tool may diminish when they must process many treated vacancies simultaneously. We investigate both possibilities by exploiting variation in the composition of caseworkers' concurrent caseloads. While the overall treatment share is fixed at 50% (75% after February 6, 2025), the realized share within a caseworker's concurrent caseload varies due to the small number of vacancies per caseworker per 7-day window.

Specifically, we measure the share of a caseworker's other preselection-mode vacancies that are assigned to treatment, using a rolling 7-day window centered on each vacancy's creation date. We then estimate how outcomes for both treated and control vacancies vary with this share, allowing us to identify both spillovers to control vacancies and congestion effects on treated vacancies.

Figure 8 displays predicted vacancy filling rates as a function of the treated share among concurrent vacancies, separately for treated (solid line) and control (dashed line) vacancies. Panel (a) shows results for all hires, while panels (b) and (c) decompose by channel.

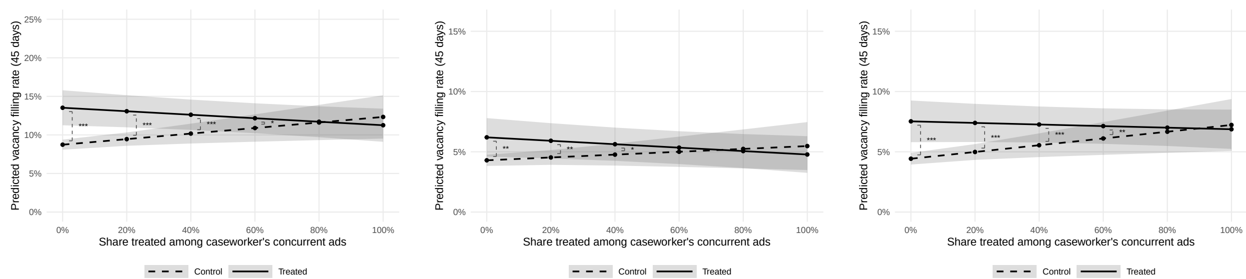
The results reveal two important patterns. First, control vacancies benefit when more of the caseworker's other vacancies are treated: the dashed line slopes upward, indicating positive spillovers. This is consistent with the interpretation that the matching tool saves caseworker time on treated vacancies, freeing up capacity that benefits control vacancies. Second, the treatment effect—the gap between solid and dashed lines—shrinks as the treated share increases. When none of a caseworker's other vacancies are treated, the treatment effect on overall hiring is approximately 5 percentage points. However, this effect falls to near zero when all other vacancies are treated.

Panel (b) shows that the sourcing channel exhibits a diminishing treatment effect as the treated share increases, for shares of 20% and lower, the treatment impact is significant at the 5%-level, while positive spillovers to control vacancies are limited. In contrast, panel (c) reveals that the screening channel drives the positive spillovers: control vacancies benefit significantly when more of the caseworker's other vacancies are treated. This suggests that time savings from the matching tool on treated vacancies free up caseworker capacity that is reallocated toward screening tasks on other vacancies.

Table 6 reports the underlying regression estimates. Results are robust at the 90-day horizon (Table A9 in the Appendix).

These findings have important implications. The positive spillovers to control vacancies—primarily through the screening channel—suggest that even partial treatment coverage can generate benefits for untreated vacancies as caseworkers reallocate time savings toward screening tasks.

Figure 7: Caseworker Congestion: Treatment Effect by Workload Composition



(a) All hires (b) Hires via sourcing (c) Hires via screening

Notes: Predicted vacancy filling rates within 45 days based on the OLS estimates reported in Table 6. The x-axis shows the share of treated vacancies among the caseworker's concurrent workload, defined as the rolling sum of preselection-mode vacancies assigned to the same caseworker within a centered 7-day window around the vacancy's creation date, excluding the vacancy itself. Predictions are computed at mean effect for all other covariates and fixed effects. Solid lines show treated vacancies; dashed lines show control vacancies. Shaded areas represent 95% confidence intervals. Significance stars indicate the statistical significance of the treatment effect. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The previous analysis revealed that the matching tool generates positive spillovers to control vacancies primarily through the screening channel, while the direct treatment effect operates mainly through sourcing. A natural question is whether these patterns vary with vacancy characteristics—in particular, whether the tool's effectiveness depends on how difficult the vacancy is to fill.

We split the sample at the median of the IEE score, France Travail's prediction of the probability of filling the vacancy within 30 days. "Easy-to-fill" vacancies are those with above-median predicted fill rates, while "hard-to-fill" vacancies have below-median predictions. We then re-estimate the spillover regressions separately for each group.

Figure 8 and Table 7 present the results. Several patterns emerge. First, at low treatment intensity—when few of the caseworker's concurrent vacancies are treated—the matching tool generates significant treatment effects across both vacancy types. However, the underlying mechanisms differ.

For easy-to-fill vacancies (left column), the treatment effect operates primarily through the screening channel. Panel (e) shows that control vacancies benefit substantially when

Table 6: Treatment Effect by Caseworker Workload Composition

	Filled within 45 days		
	(1)	(2)	(3)
Dependent Var.:	All	Sourcing	Screening
Treated	0.0479*** (0.0110)	0.0191** (0.0078)	0.0310*** (0.0084)
Concurrent ads same CW	-0.0006 (0.0007)	0.0003 (0.0005)	-0.0009* (0.0005)
Concurrent ads (preselection) same CW	-0.0019* (0.0011)	-0.0015** (0.0007)	-0.0004 (0.0008)
Share treated among concurrent ads	0.0360*** (0.0138)	0.0118 (0.0098)	0.0280*** (0.0105)
Treated × Share treated	-0.0586*** (0.0173)	-0.0261** (0.0123)	-0.0346*** (0.0131)
Strata × sub-experiment FE	X	X	X
Agency × sub-experiment FE	X	X	X
Caseworker FE	X	X	X
Observations	14,654	14,654	14,654
R2	0.07314	0.07246	0.06346
Control mean	0.1054	0.0523	0.0547
Mean concurrent ads (preselection)	4.150	4.150	4.150

Notes: OLS regressions of vacancy filling outcomes (indicator for at least one hire within 45 days) on treatment status, share of treated ads among caseworker's concurrent workload, and their interaction. One observation is one job posting. Concurrent workload is measured as the rolling sum of preselection-mode vacancies assigned to the same caseworker within a centered 7-day window around the vacancy's creation date, excluding the vacancy itself. All specifications include strata × sub-experiment, agency × sub-experiment, and caseworker fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

more of the caseworker's other vacancies are treated, with the dashed line sloping upward. This is consistent with time savings from the matching tool being reallocated toward screening tasks. Easy-to-fill vacancies typically attract more job seeker applications, making screening the binding constraint on caseworker time. There is no direct effect via sourcing (panel c) for these vacancies.

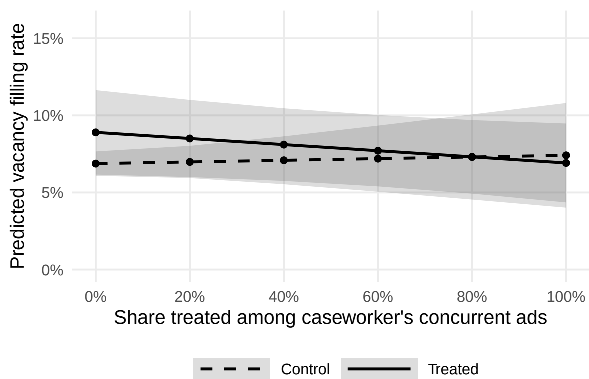
For hard-to-fill vacancies (right column), the picture is different. Panel (d) reveals a significant direct treatment effect on sourcing: treated vacancies are substantially more likely to result in a hire through caseworker-initiated referrals. This suggests that for vacancies where suitable candidates are scarce, the matching tool's recommendations provide genuine value by helping caseworkers identify candidates they would not have found otherwise. The screening channel (panel f) also shows positive effects, though the spillover pattern is less pronounced than for easy-to-fill vacancies.

These findings suggest that the matching tool operates through distinct mechanisms depending on vacancy characteristics. For easy-to-fill vacancies, where candidate supply is abundant, the tool primarily generates time savings that caseworkers reallocate toward screening. For hard-to-fill vacancies, where finding suitable candidates is the core challenge, the tool directly improves sourcing by identifying matches that caseworkers might otherwise miss. The patterns are less pronounced but similar at the 90-day horizon (Table A9).

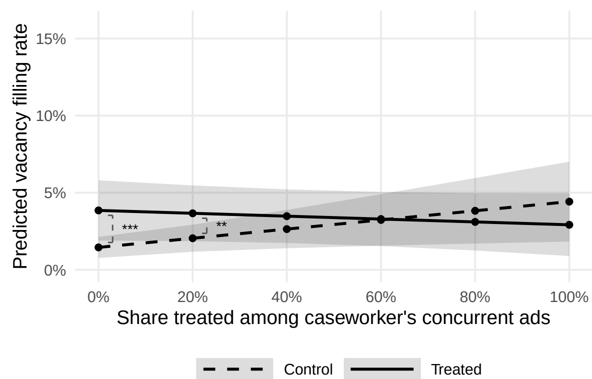
Figure 8: Predicted vacancy filling rates by treatment status and caseworker workload composition.

(a) All hires – Easy to fill

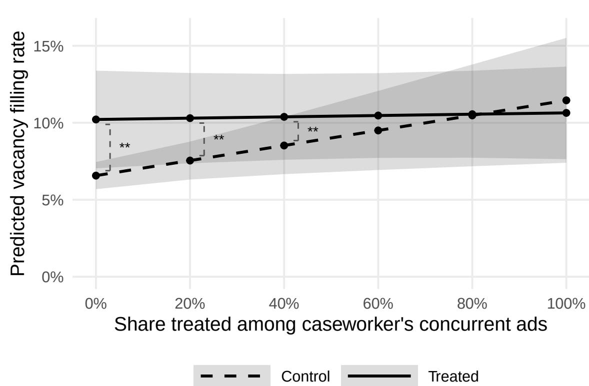
(b) All hires – Hard to fill



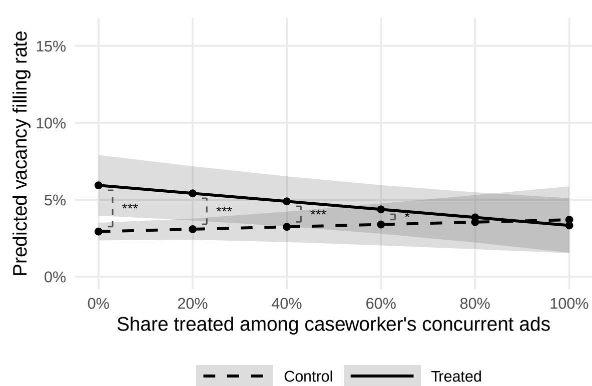
(c) Hires via sourcing – Easy to fill



(d) Hires via sourcing – Hard to fill



(e) Hires via screening – Easy to fill



(f) Hires via screening – Hard to fill

Notes: Predicted vacancy filling rates within 45 days based on the OLS estimates reported in Table 7. The x-axis shows the share of treated vacancies among the caseworker's concurrent workload, defined as the rolling sum of preselection-mode vacancies assigned to the same caseworker within a centered 7-day window around the vacancy's creation date, excluding the vacancy itself, controlling for the number of concurrent ads in the caseworkers portfolio. The average number of concurrent ads with preselection flag per 7-day window is 4.11. Predictions are computed at mean effect for all other covariates and fixed effects. "Easy to fill" and "Hard to fill" are defined by a median split on the IEE score (France Travail's prediction of the probability of filling the vacancy within 30 days). Shaded areas represent 95% confidence intervals. Significance stars indicate the statistical significance of the treatment effect: *** $p < 0.01$, ** $p < 0.05$, *

Table 7: Treatment effect by vacancy filling difficulty and caseworker workload composition (45-day horizon).

	Easy to fill			Hard to fill		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Var.:	All	Sourcing	Screening	All	Sourcing	Screening
Treated	0.0547*** (0.0193)	0.0203 (0.0134)	0.0365** (0.0155)	0.0509*** (0.0132)	0.0240*** (0.0093)	0.0300*** (0.0096)
Concurrent ads same CW	-0.0016 (0.0012)	-4.18e-5 (0.0008)	-0.0016* (0.0009)	0.0001 (0.0009)	0.0007 (0.0007)	-0.0005 (0.0006)
Conc. ads with preselection flag same CW	-0.0014 (0.0020)	-0.0011 (0.0014)	-0.0002 (0.0015)	-0.0022 (0.0014)	-0.0017* (0.0010)	-0.0004 (0.0010)
Share treated among conc. ads with flag	0.0474* (0.0243)	0.0054 (0.0167)	0.0489** (0.0197)	0.0357** (0.0162)	0.0297** (0.0124)	0.0076 (0.0109)
Treated × Share treated	-0.0686** (0.0299)	-0.0252 (0.0211)	-0.0446* (0.0239)	-0.0690*** (0.0206)	-0.0390** (0.0156)	-0.0337** (0.0143)
Strata × sub-experiment FE	X	X	X	X	X	X
Agency × sub-experiment FE	X	X	X	X	X	X
Caseworker FE	X	X	X	X	X	X
Observations	6,847	6,847	6,847	6,564	6,564	6,564
R2	0.12398	0.12458	0.11639	0.10923	0.09878	0.10250
Control mean	0.1485	0.0709	0.0806	0.0646	0.0332	0.0319
Mean concurrent ads (preselection)	4.070	4.070	4.070	4.180	4.180	4.180

Notes: OLS regressions of vacancy filling outcomes (dummy for at least one hire within 45 days) on treatment status, share of treated ads among caseworker's concurrent workload, and their interaction. Concurrent workload is measured as the rolling sum of preselection-mode vacancies assigned to the same caseworker within a centered 7-day window around the vacancy's creation date, excluding the vacancy itself. "Easy to fill" and "Hard to fill" are defined by a median split on the IEE score (France Travail's prediction of the probability of filling the vacancy within 30 days). All specifications include strata × sub-experiment, agency × sub-experiment, and caseworker fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

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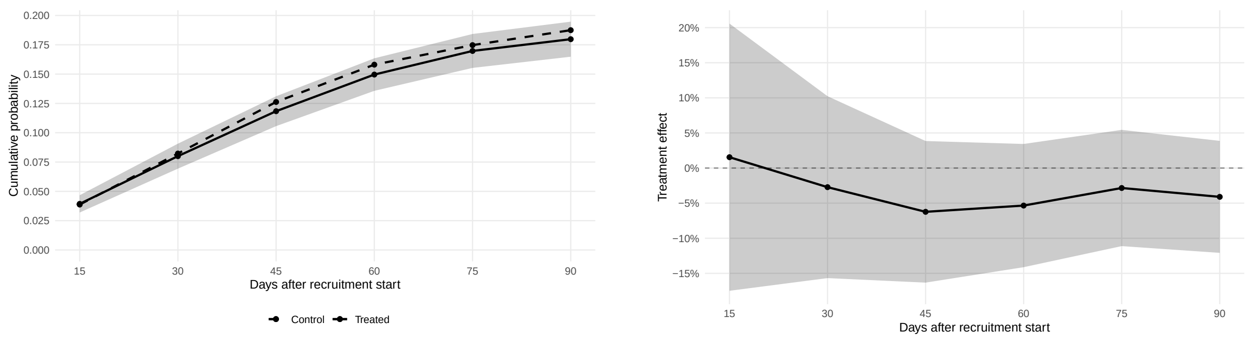
A Extra Figures

Figure A1: Placebo Test: Pre-Treatment Period



(a) Cumulative hiring probability (b) Treatment effect

Notes: Placebo test using vacancies from November 2023 to April 2024, before the experiment began. Treatment status is assigned based on the establishment's eventual treatment assignment. Panel (a) shows cumulative hiring probabilities for "treated" and "control" vacancies. Panel (b) shows the placebo treatment effect as a percentage of the control mean. Estimates are based on stratified randomization inference with strata $\times$ sub-experiment fixed effects. Shaded areas and error bars represent 95% confidence intervals.

Table A1: Robustness: Strata Fixed Effects Only

	Vacancy filled					
	(1)	(2)	(3)	(4)	(5)	(6)
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.0028 (0.0032)	0.0092* (0.0049)	0.0136** (0.0060)	0.0153** (0.0067)	0.0166** (0.0072)	0.0134* (0.0075)
Strata × sub-experiment FE	X	X	X	X	X	X
Observations	17,328	17,328	17,328	17,328	17,328	17,328
R2	0.00128	0.00199	0.00239	0.00267	0.00402	0.00464
Control mean	0.0315	0.0723	0.1091	0.1377	0.1591	0.1757

Notes: OLS regressions of vacancy filling outcomes on treatment status. Specifications include strata $\times$ sub-experiment fixed effects only (no agency fixed effects). Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A2: Robustness: Agency Fixed Effects

	Vacancy filled					
	(1)	(2)	(3)	(4)	(5)	(6)
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.0038 (0.0032)	0.0099** (0.0049)	0.0144** (0.0059)	0.0157** (0.0066)	0.0173** (0.0070)	0.0131* (0.0073)
Strata × sub-experiment FE	X	X	X	X	X	X
Agency × sub-experiment FE	X	X	X	X	X	X
Observations	17,328	17,328	17,328	17,328	17,328	17,328
R2	0.01641	0.01745	0.01843	0.02220	0.02563	0.02627
Control mean	0.0315	0.0723	0.1091	0.1377	0.1591	0.1757

Notes: OLS regressions of vacancy filling outcomes on treatment status. Specifications include strata × sub-experiment and agency × sub-experiment fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3: Robustness: Additional Controls

	Vacancy filled					
	(1)	(2)	(3)	(4)	(5)	(6)
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.0037 (0.0032)	0.0094* (0.0048)	0.0138** (0.0059)	0.0148** (0.0065)	0.0163** (0.0069)	0.0120* (0.0072)
Number of positions	0.0053* (0.0029)	0.0049 (0.0039)	0.0076 (0.0047)	0.0072 (0.0049)	0.0091* (0.0053)	0.0118** (0.0056)
Strata × sub-experiment FE	X	X	X	X	X	X
Agency × sub-experiment FE	X	X	X	X	X	X
Pre-experiment hires quintile	X	X	X	X	X	X
Observations	17,328	17,328	17,328	17,328	17,328	17,328
R2	0.01872	0.02194	0.02477	0.03097	0.03581	0.03767
Control mean	0.0315	0.0723	0.1091	0.1377	0.1591	0.1757

Notes: OLS regressions of vacancy filling outcomes on treatment status, controlling for the number of positions advertised and pre-experiment hiring quintiles. Specifications include strata × sub-experiment and agency × sub-experiment fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4: Robustness: Poisson Specification

	Number of hires via referral					
	(1)	(2)	(3)	(4)	(5)	(6)
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.1514 (0.1170)	0.1407* (0.0815)	0.1146* (0.0664)	0.0993 (0.0609)	0.0647 (0.0575)	0.0312 (0.0549)
Number of positions (log)	0.9406*** (0.1898)	0.6493*** (0.1520)	0.6880*** (0.1291)	0.6281*** (0.1194)	0.6723*** (0.1132)	0.6868*** (0.1037)
Strata × sub-experiment FE	X	X	X	X	X	X
Agency × sub-experiment FE	X	X	X	X	X	X
Observations	16,313	17,223	17,305	17,328	17,328	17,328
Control mean	0.0315	0.0723	0.1091	0.1377	0.1591	0.1757

Notes: Poisson regressions of the number of hires via referral on treatment status. Specifications include strata × sub-experiment and agency × sub-experiment fixed effects and control for the log number of positions advertised. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Robustness: All Positions Filled Outcome

	All positions filled via referral					
	(1)	(2)	(3)	(4)	(5)	(6)
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.0048 (0.0031)	0.0119** (0.0048)	0.0160*** (0.0059)	0.0174*** (0.0065)	0.0184*** (0.0069)	0.0144** (0.0072)
Strata × sub-experiment FE	X	X	X	X	X	X
Agency × sub-experiment FE	X	X	X	X	X	X
Observations	17,328	17,328	17,328	17,328	17,328	17,328
R2	0.01630	0.01739	0.01861	0.02243	0.02584	0.02642
Control mean	0.0315	0.0723	0.1091	0.1377	0.1591	0.1757

Notes: OLS regressions where the outcome is an indicator for whether all advertised positions were filled through caseworker referrals within the specified horizon. Specifications include strata × sub-experiment and agency × sub-experiment fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Caseworker Congestion: 90-Day Horizon

	Filled within 90 days		
	(1)	(2)	(3)
Dependent Var.:	All	Sourcing	Screening
Treated	0.0466*** (0.0133)	0.0232** (0.0100)	0.0275*** (0.0101)
Concurrent ads same CW	-0.0013 (0.0008)	0.0001 (0.0006)	-0.0015** (0.0006)
Concurrent ads (preselection) same CW	-0.0017 (0.0013)	-0.0017* (0.0010)	-0.0001 (0.0010)
Share treated among concurrent ads	0.0253 (0.0168)	0.0070 (0.0126)	0.0224* (0.0130)
Treated × Share treated	-0.0567*** (0.0208)	-0.0307** (0.0157)	-0.0273* (0.0157)
Strata × sub-experiment FE	X	X	X
Agency × sub-experiment FE	X	X	X
Caseworker FE	X	X	X
Observations	14,654	14,654	14,654
R2	0.07952	0.08038	0.06669
Control mean	0.1710	0.0875	0.0893
Mean concurrent ads (preselection)	4.150	4.150	4.150

Notes: OLS regressions of vacancy filling outcomes (indicator for at least one hire within 90 days) on treatment status, share of treated ads among caseworker's concurrent workload, and their interaction. Concurrent workload is measured as the rolling sum of preselection-mode vacancies assigned to the same caseworker within a centered 7-day window around the vacancy's creation date, excluding the vacancy itself. All specifications include strata × sub-experiment, agency × sub-experiment, and caseworker fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A7: Placebo Test: Pre-Treatment Period

	Vacancy filled					
	(1)	(2)	(3)	(4)	(5)	(6)
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.0019 (0.0032)	0.0037 (0.0047)	-0.0003 (0.0058)	-0.0037 (0.0065)	-0.0005 (0.0068)	-0.0029 (0.0070)
Strata × sub-experiment FE	X	X	X	X	X	X
Agency × sub-experiment FE	X	X	X	X	X	X
Observations	18,615	18,615	18,615	18,615	18,615	18,615
R2	0.01000	0.01333	0.01619	0.01602	0.01789	0.01889
Control mean	0.0387	0.0823	0.1262	0.1581	0.1747	0.1875

Notes: Placebo test using vacancies from November 2023 to April 2024, before the experiment began. Treatment status is assigned based on the establishment's eventual treatment assignment. Specifications include strata × sub-experiment and agency × sub-experiment fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Treatment Effect on Number of Candidates Presented to Firm

	Number of candidates presented to firm (all)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Horizon:	3 days	5 days	10 days	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.0516 (0.0353)	0.0461 (0.0329)	0.0447 (0.0293)	0.0438 (0.0277)	0.0469* (0.0269)	0.0401 (0.0282)	0.0331 (0.0304)	0.0468 (0.0317)	0.0301 (0.0338)
Strata x subexperiment FE	X	X	X	X	X	X	X	X	X
Unempl. office x subexperiment FE	X	X	X	X	X	X	X	X	X
Observations	16,885	16,694	16,264	15,705	13,688	11,962	10,036	8,128	6,348
Control mean	2.351	3.102	4.424	5.286	6.537	7.103	7.329	7.372	7.559

Notes: Poisson regression estimates. Outcome: number of candidates presented to the firm within the specified number of days after the vacancy enters preselection mode. One observation is one job posting. All specifications include strata × subexperiment and agency × subexperiment fixed effects. Standard errors clustered at the establishment level in parentheses. Subexperiment denotes an indicator variable for whether the preselection process for the vacancy started after February 6, 2025, when the treatment share was increased from 50% to 75%. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A9: Treatment effect by vacancy filling difficulty and caseworker workload composition (90-day horizon).

	Easy to fill			Hard to fill		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Var.:	All	Sourcing	Screening	All	Sourcing	Screening
Treated	0.0619*** (0.0217)	0.0270 (0.0165)	0.0384** (0.0175)	0.0436** (0.0182)	0.0263* (0.0137)	0.0223* (0.0131)
Concurrent ads same CW	-0.0026* (0.0013)	-8.68e-5 (0.0010)	-0.0025** (0.0011)	-0.0004 (0.0011)	-0.0001 (0.0009)	-0.0003 (0.0008)
Conc. ads with preselection flag same CW	-0.0011 (0.0023)	-0.0014 (0.0018)	-6.63e-5 (0.0017)	-0.0020 (0.0017)	-0.0013 (0.0013)	-0.0005 (0.0012)
Share treated among conc. ads with flag	0.0322 (0.0277)	-0.0047 (0.0205)	0.0449* (0.0230)	0.0323 (0.0232)	0.0256 (0.0178)	0.0059 (0.0165)
Treated × Share treated	-0.0698** (0.0339)	-0.0270 (0.0257)	-0.0395 (0.0273)	-0.0596** (0.0287)	-0.0417* (0.0222)	-0.0241 (0.0203)
Strata × sub-experiment FE	X	X	X	X	X	X
Agency × sub-experiment FE	X	X	X	X	X	X
Caseworker FE	X	X	X	X	X	X
Observations	6,847	6,847	6,847	6,564	6,564	6,564
R2	0.13557	0.12561	0.12497	0.12345	0.12910	0.10802
Control mean	0.2168	0.1071	0.1177	0.1297	0.0687	0.0651
Mean concurrent ads (preselection)	4.070	4.070	4.070	4.180	4.180	4.180

Notes: OLS regressions of vacancy filling outcomes (dummy for at least one hire within 90 days) on treatment status, share of treated ads among caseworker's concurrent workload, and their interaction. Concurrent workload is measured as the rolling sum of preselection-mode vacancies assigned to the same caseworker within a centered 7-day window around the vacancy's creation date, excluding the vacancy itself. "Easy to fill" and "Hard to fill" are defined by a median split on the IEE score (France Travail's prediction of the probability of filling the vacancy within 30 days). All specifications include strata × sub-experiment, agency × sub-experiment, and caseworker fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.